Optimal Reconfiguration of Systems in Aeronautics

Lecture 13: "Robust Optimal Control and Admissible Model Matching"

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Outline

1. Optimal Model Matching
2. Modified Pseudo-Inverse Method – Revisited
3. B747 Application Example

Outline

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Preliminaries I

This section provides some results on robust stability bounds.

It is useful when the modified pseudo-inverse is used for the stabilization of the unstable approximate control obtained with the pseudo-inverse in the Exact Model Matching and when the robust optimal control is used for the robust stabilization and the optimal matching of the ideal reference model.

Robust stability for modified pseudo-inverse method I

Theorem (Zhou and Khargonekar; 1987)

Consider the post-fault closed-loop system

$$\dot{x}(t) = \hat{A}_f x(t) \equiv (\hat{A}_n^* + E_f)x(t)\,, \tag{1}$$

where $\hat{A}_f, \hat{A}_n^* \in \mathbb{R}^{n\times n}$ *are the post-fault closed-loop matrix and the ideal reference model, respectively, and* $E_f = \hat{A}_f - \hat{A}_n^*$ *represents the difference between the two closed-loop matrices, in particular assumed to be in the following form*

$$E_f = \sum_{i=1}^{m} c_{f_i} E_{f_i}\,, \tag{2}$$

where E_{f_i} *are constant matrices and coefficients* c_{f_i} *vary in intervals around zero.*

Robust stability for modified pseudo-inverse method II

Theorem (Zhou and Khargonekar; 1987 – cont'd)

Moreover, let $\hat{X}_f$ *be the solution of the Lyapunov equation*

$$\hat{A}_f^\top \hat{X}_f + \hat{X}_f \hat{A}_f + 2I_n = 0\,, \tag{3}$$

and define $\hat{X}_{f_i}$, $\hat{X}_{f_e}$ *as*

$$\hat{X}_{f_i} = (E_{f_i}^\top \hat{X}_f + \hat{X}_f E_{f_i})/2 = [\hat{X}_f E_{f_i}]_S\,, \ \textit{for } i = 1, 2, \ldots, m\,, \tag{4}$$

$$\hat{X}_{f_e} = \begin{bmatrix} \hat{X}_{f_1} & \hat{X}_{f_2} & \ldots & \hat{X}_{f_m} \end{bmatrix}\,, \tag{5}$$

where $[X]_S$ *denotes the symmetric-part of the matrix-argument, i.e.,* $[X]_S = \left(X + X^\top\right)/2$.

Robust stability for modified pseudo-inverse method III

Theorem (Zhou and Khargonekar; 1987 – cont'd)

Then $\hat{A}_f$ *is stable if:*

$$\sum_{i=1}^{m} c_{f_i}^2 < \frac{1}{\sigma_{\max}^2(\hat{X}_{f_e})}\,, \quad \textit{or} \tag{6a}$$

$$\sum_{i=1}^{m} |c_{f_i}|\sigma_{\max}(\hat{X}_{f_i}) < 1\,, \quad \textit{or} \tag{6b}$$

$$|c_{f_j}| < \frac{1}{\sigma_{\max}\left(\sum_{i=1}^{m} |\hat{X}_{f_i}|\right)}\,, \quad \textit{for } j = 1, 2, \ldots, m\,, \tag{6c}$$

where $|X|$ *indicates the matrix-argument with its elements in absolute value and* $\sigma_{\max}$ *stands for the maximum singular value.* □

Robust stability on nonlinear perturbations I

Consider the post-fault system

$$\dot{x}(t) = A_f x(t) + B_f u_f(t) + \Gamma_f[x(t), u_f(t)]\,, \tag{7}$$

where $A_f \in \mathbb{R}^{n\times n}, B_f \in \mathbb{R}^{n\times m}$ denote the post-fault pair of system matrices and Γ_f is a nonlinear vector function representing modeling errors or parameter variations in the system model (A_f, B_f).

In optimal model matching control design, the state-feedback is chosen to minimize the performance index

$$J_f(\alpha_f, \hat{A}_n^*) = \int_0^{+\infty} e^{2\alpha_f t}\left(\left\|\dot{x}(t) - \hat{A}_n^* x(t)\right\|_2^2 + \left\|\hat{A}_n^* x(t)\right\|_2^2\right) dt\,, \tag{8}$$

where $\hat{A}_n^* \in \mathbb{R}^{n\times n}$ is the nominal closed-loop ideal model, with $|\lambda(\hat{A}_n^*)| \geq \alpha_f$ the absolute value of the real-part of any eigenvalue of the ideal model matrix.

Robust stability on nonlinear perturbations II

In this respect, the optimal control $u_f(t)$, which minimizes $J_f(\alpha_f, \hat{A}_n^*)$ subject to (7), is

$$u_f(t) = -K_f x(t)\,, \text{ with } K_f = R_f^{-1}(B_f^\top X_f + L_f)\,, \tag{9}$$

where matrix $X_f \in \mathbb{R}^{n\times n}$ is the unique stabilizing symmetric positive-definite solution of the generalized continuous algebraic Riccati equation (GCARE)

$$GCARE_f:\ \tilde{A}_f^\top X_f + X_f\tilde{A}_f - X_f B_f R_f^{-1} B_f^\top X_f + \tilde{Q}_f = 0\,, \tag{10}$$

with

$$\tilde{A}_f = A_f + \alpha_f I_n - B_f R_f^{-1} L_f\,, \tag{11a}$$

$$\tilde{Q}_f = Q_f - L_f^\top R_f^{-1} L_f\,, \tag{11b}$$

for

$$Q_f = A_f^\top A_f - A_f^\top \hat{A}_n^* - \hat{A}_n^{*\,T} A_f + 2\hat{A}_n^{*\,T}\hat{A}_n^*\,, \tag{12a}$$

$$R_f = B_f^\top B_f\,, \tag{12b}$$

$$L_f = B_f^\top (A_f - \hat{A}_n^*)\,. \tag{12c}$$

Robust stability on nonlinear perturbations III

The proof of this result is being done by reducing the performance index (8) to the generalized-form of the performance index

$$J_f = \int_0^{+\infty} e^{2\alpha_f t} \left(\|x(t)\|_{Q_f}^2 + 2u_f^\top(t) L_f x(t) + \|u_f(t)\|_{R_f}^2 \right) dt\,. \tag{13}$$

For the results that follow, let $\|W\|_S$, $\|W\|_E$, $\kappa(W)$, $\lambda_{\max}(W)$, and $\lambda_{\min}(W)$ denote the spectral-norm, Euclidean-norm, condition-number, maximum- and minimum- eigenvalue, respectively, of the matrix-argument W, where $\mathrm{tr}\,(W)$ indicates its trace.

Robust stability on nonlinear perturbations IV

Theorem (Patel, Toda, and Sridhar; 1977)

Let

$$T_f \triangleq \tilde{A}_f^\top X_f + X_f \tilde{A}_f + 2\tilde{Q}_f. \tag{14}$$

If the nonlinear vector function $\Gamma_f[z_f(t)]$ *satisfies the condition*

$$\begin{aligned}\frac{\|\Gamma_f[z_f(t)]\|_E}{\|z_f(t)\|_E} \leq \mu_f &= \frac{1}{2}\frac{1}{\left\|T_f^{-1}\right\|_S \|X_f\|_S} + \frac{\alpha_f}{\kappa(X_f)} \\ &= \frac{1}{2}\frac{1}{\left\|T_f^{-1}\right\|_S \|X_f\|_S} + \frac{\alpha_f}{\left\|X_f^{-1}\right\|_S \|X_f\|_S} \\ &= \frac{1}{2}\frac{\lambda_{\min}(T_f)}{\lambda_{\max}(X_f)} + \alpha_f \frac{\lambda_{\min}(X_f)}{\lambda_{\max}(X_f)}\,,\end{aligned} \tag{15}$$

Robust stability on nonlinear perturbations V

Theorem (Patel, Toda, and Sridhar; 1977 – cont'd)

then the closed-loop system

$$\dot{x}(t) = \left[A_f - B_f R_f^{-1}\left(B_f^\top X_f + L_f\right)\right] x(t) + \Gamma_f[z_f(t)] \tag{16}$$

is stable, where $\kappa(X_f)$ $(\equiv \|X_f\|_S \left\|X_f^{-1}\right\|_S)$ *denotes the condition-number of* X_f*, and* $z_f(t)$ *generally expresses the effects of* $x(t)$ *and* $u_f(t)$ *through* Γ_f *onto the system.* $\square$

Robust stability on nonlinear perturbations VI

Remark

Comparing (14) to the post-fault GCARE (10) yields

$$T_f = \tilde{Q}_f + X_f B_f R_f^{-1} B_f^\top X_f, \tag{17}$$

and, since $\tilde{Q}_f > 0$, $X_f B_f R_f^{-1} B_f^\top X_f \geq 0$, it follows that $T_f > 0$, therefore T_f is a symmetric positive-definite matrix; hence, condition (15) bounds $\|\Gamma_f[z_f(t)]\|_E / \|z_f(t)\|_E$ by some positive number; also, since $T_f - \tilde{Q}_f \geq 0$, then $\lambda_{\min}(T_f) \geq \lambda_{\min}(\tilde{Q}_f) > 0$, and $\left\| T_f^{-1} \right\|_S$ can be replaced by $\left\| \tilde{Q}_f^{-1} \right\|_S$ in (15). $\square$

Robust stability on linear perturbations I

Consider that the perturbation $\Gamma_f[x(t), u_f(t)]$ is linear in $x(t)$ and $u_f(t)$.

Therefore

$$\Gamma_f[x(t), u_f(t)] = \Delta A_f x(t) + \Delta B_f u_f(t)\,, \tag{18}$$

where $\Delta A_f \in \mathbb{R}^{n\times n}$ and $\Delta B_f \in \mathbb{R}^{n\times m}$ are constant matrices.

Robust stability on linear perturbations II

System (7) therefore becomes

$$\dot{x}(t) = (A_f + \Delta A_f)x(t) + (B_f + \Delta B_f)u_f(t)\,, \tag{19}$$

for which the state-feedback control law is given by

$$u_f(t) = -K_f x(t)\,, \text{ with } K_f = R_f^{-1}(B_f^\top X_f + L_f)\,, \tag{20}$$

and hence (18) becomes

$$\Gamma_f[z_f(t)] = (\Delta A_f + \Delta B_f K_f)x(t)\,. \tag{21}$$

The following theorems give bounds on ΔA_f and ΔB_f for the stability of the closed-loop system, where $\Gamma_f[z_f(t)]$ is given by (21), i.e., of the system

$$\dot{x}(t) = \left[(A_f + \Delta A_f) - (B_f + \Delta B_f)K_f\right]x(t)\,. \tag{22}$$

Robust stability on linear perturbations III

Theorem (Patel, Toda, and Sridhar; 1977)

If the perturbation matrices ΔA_f and ΔB_f satisfy the condition

$$\|\Delta A_f\|_S + \|\Delta B_f\|_S \|K_f\|_S \leq \mu_f = \frac{1}{2} \frac{1}{\left\| T_f^{-1} \right\|_S \|X_f\|_S} + \frac{\alpha_f}{\left\| X_f^{-1} \right\|_S \|X_f\|_S}, \tag{23}$$

then the closed-loop system (22) is stable. □

Robust stability on linear perturbations IV

Theorem (Patel, Toda, and Sridhar; 1977)

If the perturbation matrices ΔA_f and ΔB_f satisfy the condition

$$\|\Delta A_f\|_E + \|\Delta B_f\|_E \|K_f\|_S \leq \mu_f = \frac{1}{2} \frac{1}{\left\| T_f^{-1} \right\|_S \|X_f\|_S} + \frac{\alpha_f}{\left\| X_f^{-1} \right\|_S \|X_f\|_S}, \tag{24}$$

then the closed-loop system (22) is stable. □

Robust stability on linear perturbations V

Theorem (Patel, Toda, and Sridhar; 1977)

Let $\epsilon_{\Delta A_f}$ and $\epsilon_{\Delta B_f}$ denote some given bounds on all elements of the perturbation matrices ΔA_f and ΔB_f, respectively.

Then, if

$$\epsilon_{\Delta A_f} + \sqrt{\frac{m}{n}} \|K_f\|_S \epsilon_{\Delta B_f} \leq \frac{1}{n}\mu_f = \frac{1}{n}\left(\frac{1}{2}\frac{1}{\left\|T_f^{-1}\right\|_S \|X_f\|_S} + \frac{\alpha_f}{\left\|X_f^{-1}\right\|_S \|X_f\|_S}\right), \tag{25}$$

then the closed-loop system (22) is stable. □

Classical robust optimal model matching I

While previously admissibility was defined using only algebraic constraints, in order to develop an extension to the classical modified pseudo-inverse method for model matching, in the sequel the criterion is shifted to an integral cost.

In this respect, let $\hat{A}_n^*$ be the optimal nominal closed-loop matrix with $|\lambda(\hat{A}_n^*)| \geq \alpha_f$, where $|\lambda(\hat{A}_n^*)|$ is the maximum absolute value of the real-part of the eigenvalues of $\hat{A}_n^*$; the exponent α_f is defined as the desired minimal decay-rate of the post-fault closed-loop system.

Then, the post-fault state-feedback is chosen to minimize the classical robust optimal model matching performance index

$$J_{f_c}(\alpha_f, \hat{A}_n^*) = \int_{t_f}^{+\infty} e^{2\alpha_f t} \left(\left\| \dot{x}(t) - \hat{A}_n^* x(t) \right\|_2^2 + \left\| \hat{A}_n^* x(t) \right\|_2^2 \right) dt\,; \tag{26}$$

such a performance integral represents the requirement that the post-fault system optimally matches the model as specified by $\hat{A}_n^*$.

Classical robust optimal model matching II

The optimal control $u_{f_c}(t)$ which minimizes $J_{f_c}(\alpha_f, \hat{A}_n^*)$ subject to (A_f, B_f) is shown to be

$$u_{f_c}(t) = -K_{f_c} x(t)\,, \text{ with } K_{f_c} = R_{f_c}^{-1}(B_f^\top X_{f_c} + L_{f_c})\,, \tag{27}$$

where matrix X_{f_c} is the unique stabilizing symmetric positive-semidefinite solution of the generalized continuous algebraic Riccati equation (GCARE)

$$GCARE_{f_c} \;:\; \tilde{A}_{f_c}^\top X_{f_c} + X_{f_c}\tilde{A}_{f_c} - X_{f_c} B_f R_{f_c}^{-1} B_f^\top X_{f_c} + \tilde{Q}_{f_c} = 0\,, \tag{28}$$

with

$$\tilde{A}_{f_c} = A_f + \alpha_f I_n - B_f R_{f_c}^{-1} L_{f_c}\,, \tag{29a}$$

$$\tilde{Q}_{f_c} = Q_{f_c} - L_{f_c}^\top R_{f_c}^{-1} L_{f_c}\,; \tag{29b}$$

Classical robust optimal model matching III

the required post-fault weighting matrices in (28) respectively (29) are derived from the nominal and post-fault closed-loop systems matching, being computed as

$$Q_{f_c} = A_f^\top A_f - A_f^\top \hat{A}_n^* - \hat{A}_n^{*\,T} A_f + 2\hat{A}_n^{*\,T} \hat{A}_n^* \,, \tag{30a}$$

$$R_{f_c} = B_f^\top B_f \,, \tag{30b}$$

$$L_{f_c} = B_f^\top (A_f - \hat{A}_n^*) \,, \tag{30c}$$

thus avoiding any trial-and-error time-consuming procedure to finding Q_{f_c}, R_{f_c}, and L_{f_c}, in real-time operation.

To prove the above result, the performance integral (26) is reduced to a corresponding generalized-form

$$J_f(x_f, u_f) = \int_{t_f}^{+\infty} e^{2\alpha_f t} \left(\|x(t)\|_{Q_f}^2 + 2u_f^\top(t) L_f x(t) + \|u_f(t)\|_{R_f}^2 \right) dt \,. \tag{31}$$

Classical robust optimal model matching IV

Starting from the integrand,

$$
\begin{aligned}
&\left\|\dot{x}(t) - \hat{A}_n^* x(t)\right\|_2^2 + \left\|\hat{A}_n^* x(t)\right\|_2^2 = \\
=&\left\|A_f x(t) - \hat{A}_n^* x(t) - B_f K_{f_c} x(t)\right\|_2^2 + \left\|\hat{A}_n^* x(t)\right\|_2^2 \\
=&x^\top(t)\left[(A_f - \hat{A}_n^*) - B_f K_{f_c}\right]^\top \left[(A_f - \hat{A}_n^*) - B_f K_{f_c}\right] x(t) + x^\top(t)\hat{A}_n^{*\,T}\hat{A}_n^* x(t) \\
=&x^\top(t)(A_f - \hat{A}_n^*)^\top(A_f - \hat{A}_n^*)x(t) + x^\top(t)(A_f - \hat{A}_n^*)^\top(-B_f K_{f_c})x(t) \\
&+ x^\top(t)(-B_f K_{f_c})^\top(A_f - \hat{A}_n^*)x(t) + x^\top(t)(-B_f K_{f_c})^\top(-B_f K_{f_c})x(t) \\
&+ x^\top(t)\hat{A}_n^{*\,T}\hat{A}_n^* x(t) \\
=&x^\top(t)\left(A_f^\top A_f - A_f^\top \hat{A}_n^* - \hat{A}_n^{*\,T} A_f + 2\hat{A}_n^{*\,T}\hat{A}_n^*\right)x(t) + u_{f_c}^\top(t)\left(B_f^\top B_f\right)u_{f_c}(t) \\
&+ 2u_{f_c}^\top(t)\left[B_f^\top(A_f - \hat{A}_n^*)\right]x(t) \\
=&\|x(t)\|_{Q_{f_c}}^2 + 2u_{f_c}^\top(t)L_{f_c}x(t) + \left\|u_{f_c}(t)\right\|_{R_{f_c}}^2,
\end{aligned}
\tag{32}
$$

matrices $Q_{f_c}, R_{f_c}, L_{f_c}$ are identified exactly as in (30).

Post-fault reconfiguration with prescribed degree-of-stability I

To probe the supplementary effect on the closed-loop eigenvalues of imposing the stability-margin constraint α_f on the integral criterion (31), proceed by setting

$$\hat{x}(t) = e^{\alpha_f t} x(t)\,, \tag{33a}$$

$$\hat{u}_{f_c}(t) = e^{\alpha_f t} u_{f_c}(t)\,, \tag{33b}$$

and take the derivative of (33a), respectively

$$\begin{aligned} \dot{\hat{x}}(t) &= \frac{d}{dt}\left[e^{\alpha_f t} x(t)\right] \\ &= \alpha_f e^{\alpha_f t} x(t) + e^{\alpha_f t} \dot{x}(t)\,, \end{aligned} \tag{34}$$

Post-fault reconfiguration with prescribed degree-of-stability II

which, after replacing the expression of $\dot{x}(t)$ from (35), respectively

$$M_f : \ \dot{x}(t) = A_f x(t) + B_f u_f(t) \,, \tag{35}$$

into (34), gives

$$\begin{aligned} \dot{\hat{x}}(t) &= \alpha_f \hat{x}(t) + e^{\alpha_f t} A_f x(t) + e^{\alpha_f t} B_f u_{f_c}(t) \\ &= (A_f + \alpha_f I_n)\hat{x}(t) + B_f \hat{u}_{f_c}(t) \,, \end{aligned} \tag{36}$$

under the requirement for identity of the initial conditions of both systems, precisely $(\hat{x}_f \triangleq)\, \hat{x}(t_f) = e^{\alpha_f t_f} x(t_f)$.

Post-fault reconfiguration with prescribed degree-of-stability III

Furthermore, evaluate the quadratic quantity

$$e^{2\alpha_f t}\left[x^\top(t)Q_{f_c}x(t) + 2u_{f_c}^\top(t)L_{f_c}x(t) + u_{f_c}^\top(t)R_{f_c}u_{f_c}(t)\right] = \hat{x}^\top(t)Q_{f_c}\hat{x}(t) + 2\hat{u}_f^\top(t)L_{f_c}\hat{x}(t) + \hat{u}_f^\top(t)R_{f_c}\hat{u}_f(t) \triangleq \hat{J}_{f_c}(\hat{x}, \hat{u}_f, t; Q_{f_c}, R_{f_c}, L_{f_c}), \tag{37}$$

and thus see that the minimization of (31) w.r.t. (35) is equivalent to minimizing

$$\hat{J}_{f_c}(\hat{x}, \hat{u}_f, t) = \int_{t_f}^{+\infty} \hat{J}_{f_c}(\hat{x}, \hat{u}_f, t; Q_{f_c}, R_{f_c}, S_{f_c})\, dt, \tag{38}$$

with regard to

$$\dot{\hat{x}}(t) = (A_f + \alpha_f I_n)\hat{x}(t) + B_f\hat{u}_{f_c}(t); \tag{39}$$

in this respect, the optimal control $\hat{u}_{f_c}(t)$ for (38)-(39) is exactly $u_{f_c}(t)$ from (27) computed based on (28) respectively (29).

Post-fault reconfiguration with prescribed degree-of-stability IV

Moreover, by replacing in (27) the expressions for $\hat{x}(t)$ and $\hat{u}_{f_c}(t)$ from (33) yields

$$\begin{aligned} \hat{u}_{f_c}(t) &= -K_{f_c}\hat{x}(t) \;\Rightarrow \\ e^{\alpha_f t}u_{f_c}(t) &= -K_{f_c}e^{\alpha_f t}x(t) \;\Rightarrow \\ u_{f_c}(t) &= -K_{f_c}x(t)\,, \end{aligned} \tag{40}$$

from where it follows that K_{f_c} obtained from (27) respectively (28)-(29) is also the post-fault optimal control matrix for (35).

To conclude, the closed-loop system for (39) is

$$\dot{\hat{x}}(t) = (A_f + \alpha_f I_n - B_f K_{f_c})\hat{x}(t)\,, \tag{41}$$

and, since (41) is asymptotically stable under the classical stabilizability and detectability assumptions, meaning that the closed-loop eigenvalues of this system are in the stable left half-plane, that is they have negative real-parts, then the closed-loop eigenvalues for (35) under the same control of K_{f_c}, that is of $A_f - B_f K_{f_c}$, which are less by α_f than the eigenvalues of $A_f + \alpha_f I_n - B_f K_{f_c}$, all possess real-parts less than $-\alpha_f$, from where the prescribed degree-of-stability.

Post-fault reconfiguration with prescribed degree-of-stability V

Moreover, from (33a), one has

$$\begin{aligned} \hat{x}(t) &= e^{\alpha_f t} x(t) \Rightarrow \\ x(t) &= e^{-\alpha_f t} \hat{x}(t) \,, \end{aligned} \tag{42}$$

which gives for α_f the term decay-rate of the post-fault closed-loop system.

Simplified Robust Optimal Model Matching I

Recall the performance index J_{f_c} from (26) and consider its simplified version

$$J_{f_s}(\alpha_f, \hat{A}_n^*) = \int_{t_f}^{+\infty} e^{2\alpha_f t} \left(\left\| \dot{x}(t) - \hat{A}_n^* x(t) \right\|_2^2 \right) dt \,. \tag{43}$$

Then, the post-fault state-feedback $u_{f_s}(t)$, chosen to minimize $J_{f_s}(\alpha_f, \hat{A}_n^*)$ in (43) subject to (A_f, B_f), is similar to (27), that is

$$u_{f_s}(t) = -K_{f_s} x(t) \,, \text{ with } K_{f_s} = R_{f_s}^{-1}(B_f^\top X_{f_s} + L_{f_s}) \,, \tag{44}$$

with the only difference that X_{f_s} is the solution to another GCARE

$$GCARE_{f_s} \; : \; \tilde{A}_{f_s}^\top X_{f_s} + X_{f_s} \tilde{A}_{f_s} - X_{f_s} B_f R_{f_s}^{-1} B_f^\top X_{f_s} + \tilde{Q}_{f_s} = 0 \,, \tag{45}$$

Simplified Robust Optimal Model Matching II

for $\tilde{A}_{f_s}$ as in (29a), namely $\tilde{A}_{f_s} = \tilde{A}_{f_c}$, but with

$$\tilde{Q}_{f_s} = Q_{f_s} - L_{f_s}^\top R_{f_s}^{-1} L_{f_s} \,, \tag{46}$$

where the required post-fault weighting matrices R_{f_s} and S_{f_s} in (45) and (46) are the same as in (30a) and (30b), precisely $R_{f_s} = R_{f_c}$ and $S_{f_s} = S_{f_c}$, but Q_{f_s} is derived from the nominal and post-fault closed-loop Simplified Robust Optimal Model Matching (SROMM), being computed as

$$\begin{aligned} Q_{f_s} &= A_f^\top A_f - A_f^\top \hat{A}_n^* - \hat{A}_n^{*\,T} A_f + \hat{A}_n^{*\,T} \hat{A}_n^* \\ &= Q_{f_c} - \hat{A}_n^{*\,T} \hat{A}_n^* \,; \end{aligned} \tag{47}$$

again, to prove the above result, the performance integral (43) is reduced to the corresponding generalized-form (31).

Simplified Robust Optimal Model Matching III

Computing as above, start from the integrand,

$$
\begin{aligned}
&\left\|\dot{x}(t) - \hat{A}_n^* x(t)\right\|_2^2 = \\
=&\left\|A_f x(t) - \hat{A}_n^* x(t) - B_f K_{f_s} x(t)\right\|_2^2 \\
=&x^\top(t)\left[(A_f - \hat{A}_n^*) - B_f K_{f_s}\right]^\top \left[(A_f - \hat{A}_n^*) - B_f K_{f_s}\right] x(t) \\
=&x^\top(t)(A_f - \hat{A}_n^*)^\top (A_f - \hat{A}_n^*)x(t) + x^\top(t)(A_f - \hat{A}_n^*)^\top(-B_f K_{f_s})x(t) \\
&+ x^\top(t)(-B_f K_{f_s})^\top (A_f - \hat{A}_n^*)x(t) + x^\top(t)(-B_f K_{f_s})^\top(-B_f K_{f_s})x(t) \\
=&x^\top(t)\left(A_f^\top A_f - A_f^\top \hat{A}_n^* - \hat{A}_n^{*\,T} A_f + \hat{A}_n^{*\,T}\hat{A}_n^*\right) x(t) + u_{f_s}^\top(t)\left(B_f^\top B_f\right) u_{f_s}(t) \\
&+ 2u_{f_s}^\top(t)\left[B_f^\top (A_f - \hat{A}_n^*)\right] x(t) \\
=&\|x(t)\|_{Q_{f_s}}^2 + 2u_{f_s}^\top(t) L_{f_s} x(t) + \left\|u_{f_s}(t)\right\|_{R_{f_s}}^2 ,
\end{aligned}
\tag{48}
$$

matrices S_{f_s}, R_{f_s} keeping their previous form from (30) but matrix Q_{f_s} being identified exactly as in (47).

Extended Robust Optimal Model Matching I

Focus again on the performance index from (26) and consider its extended version

$$J_{f_e}(\alpha_f, \hat{A}_n^*) = \int_{t_f}^{+\infty} e^{2\alpha_f t} \left(\left\| \dot{x}(t) - \hat{A}_n^* x(t) \right\|_2^2 + \left\| u_{f_e}(t) - u_n(t) \right\|_2^2 \right) dt \,. \tag{49}$$

Then, the post-fault state-feedback $u_{f_e}(t)$, chosen to minimize $J_{f_e}(\alpha_f, \hat{A}_n^*)$ in (49) subject to (A_f, B_f), is similar to (27), that is

$$u_{f_e}(t) = -K_{f_e} x(t) \,, \text{ with } K_{f_e} = R_{f_e}^{-1} (B_f^\top X_{f_e} + L_{f_e}) \,, \tag{50}$$

with the only difference that X_{f_e} is the solution to another GCARE

$$GCARE_{f_e} \; : \; \tilde{A}_{f_e}^\top X_{f_e} + X_{f_e} \tilde{A}_{f_e} - X_{f_e} B_f R_{f_e}^{-1} B_f^\top X_{f_e} + \tilde{Q}_{f_e} = 0 \,, \tag{51}$$

Extended Robust Optimal Model Matching II

for

$$\tilde{A}_{f_e} = A_f + \alpha_f I_n - B_f R_{f_e}^{-1} L_{f_e} \text{ and} \tag{52a}$$

$$\tilde{Q}_{f_e} = Q_{f_e} - L_{f_e}^\top R_{f_e}^{-1} L_{f_e}\,, \tag{52b}$$

with

$$\begin{aligned} Q_{f_e} &= A_f^\top A_f - A_f^\top \hat{A}_n^* - \hat{A}_n^{*\,T} A_f + \hat{A}_n^{*\,T} \hat{A}_n^* + K_n^\top K_n \\ &= Q_{f_s} + K_n^\top K_n\,, \end{aligned} \tag{53a}$$

$$\begin{aligned} R_{f_e} &= B_f^\top B_f + I_m \\ &= R_{f_s} + I_m\,, \end{aligned} \tag{53b}$$

$$\begin{aligned} L_{f_e} &= B_f^\top (A_f - \hat{A}_n^*) + K_n \\ &= L_{f_s} + K_n\,; \end{aligned} \tag{53c}$$

once again, to prove the above result, the performance integral (49) is reduced to the corresponding generalized-form (31).

Extended Robust Optimal Model Matching III

Proceeding similarly as before, start from the integrand,

$$
\begin{aligned}
&\left\|\dot{x}(t) - \hat{A}_n^* x(t)\right\|_2^2 + \|u_f(t) - u_n(t)\|_2^2 = \\
=&\left\|A_f x(t) - \hat{A}_n^* x(t) - B_f K_{f_e} x(t)\right\|_2^2 + \left\|(K_{f_e} - K_n)x(t)\right\|_2^2 \\
=&x^\top(t)\left[(A_f - \hat{A}_n^*) - B_f K_{f_e}\right]^\top \left[(A_f - \hat{A}_n^*) - B_f K_{f_e}\right] x(t) \\
&+ x^\top(t)(K_{f_e} - K_n)^\top (K_{f_e} - K_n)x(t) \\
=&x^\top(t)(A_f - \hat{A}_n^*)^\top (A_f - \hat{A}_n^*)x(t) + x^\top(t)(A_f - \hat{A}_n^*)^\top (-B_f K_{f_e})x(t) \\
&+ x^\top(t)(-B_f K_{f_e})^\top (A_f - \hat{A}_n^*)x(t) + x^\top(t)(-B_f K_{f_e})^\top (-B_f K_{f_e})x(t) \\
&+ x^\top(t)(K_n^\top K_n)x(t) + x^\top(t)(K_{f_e}^\top K_{f_e})x(t) + x^\top(t)(-K_{f_e}^\top K_n)x(t) \\
&+ x^\top(t)(-K_n^\top K_{f_e})x(t) \\
=&x^\top(t)\left(A_f^\top A_f - A_f^\top \hat{A}_n^* - \hat{A}_n^{*\,T} A_f + \hat{A}_n^{*\,T}\hat{A}_n^* + K_n^\top K_n\right) x(t) \\
&+ u_{f_e}^\top(t)\left(B_f^\top B_f + I_m\right) u_{f_e}(t) + 2u_{f_e}^\top(t)\left[B_f^\top (A_f - \hat{A}_n^*) + K_n\right] x(t) \\
=&\|x(t)\|_{Q_{f_e}}^2 + 2u_{f_e}^\top(t) L_{f_e} x(t) + \left\|u_{f_e}(t)\right\|_{R_{f_e}}^2 ,
\end{aligned}
\tag{54}
$$

matrices $Q_{f_e}, R_{f_e}, L_{f_e}$ being identified exactly as in (53).

Comment on the Robust Stability Bound I

The expressions for the post-fault control-accommodation matrices from the three situations (27), (44), and (50), respectively, are valid only if certain norm-conditions between the post-fault pair of system matrices respectively these post-fault control matrices are satisfied, that is whether the post-fault situation is of the following type.

Precisely, if bounded matrices ΔA_f and ΔB_f, hopefully available from the FDIE module according to

$$A_f = A_n \pm \Delta A_f\,, \text{ respectively}$$
$$B_f = B_n \pm \Delta B_f\,,$$

satisfy one of the following conditions

$$\|\Delta A_f\|_S + \|\Delta B_f\|_S \|K_{f_k}\|_S \leq \mu_{f_k}\,, \tag{55a}$$

or

$$\|\Delta A_f\|_E + \|\Delta B_f\|_E \|K_{f_k}\|_S \leq \mu_{f_k}\,, \tag{55b}$$

or

$$\epsilon_{\Delta A_f} + \sqrt{\frac{m}{n}} \|K_{f_k}\|_S \epsilon_{\Delta B_f} \leq \frac{1}{n}\mu_{f_k}\,, \tag{55c}$$

Comment on the Robust Stability Bound II

with

$$\mu_{f_k} = \frac{1}{2}\frac{1}{\left\|T_{f_k}^{-1}\right\|_S \|X_{f_k}\|_S} + \alpha_f \frac{1}{\left\|X_{f_k}^{-1}\right\|_S \|X_{f_k}\|_S}\,, \tag{56}$$

then the impaired closed-loop system $\hat{A}_{f_k} = A_f - B_f K_{f_k}$ is stable, with

$$T_{f_k} \triangleq \tilde{A}_{f_k}^\top X_{f_k} + X_{f_k}\tilde{A}_{f_k} + 2\tilde{Q}_{f_k}\,, \tag{57}$$

for $k \in \{c, s, e\}$, where $\|X\|_S$ stands for the spectral-norm of the matrix-argument X, while $\epsilon_{\Delta A_f}$ and $\epsilon_{\Delta B_f}$ denote estimated bounds on all elements of the impairment matrices ΔA_f and ΔB_f, respectively.

The proofs of (55a), (55b), and (55c), respectively, are derived from the fact that the dissipation-function $\dot{V}_{f_k}[x(t)] \leq 0$, $\forall\, x(t)$, when $\alpha_f \geq 0$, for the Lyapunov energy-function $V_{f_k}[x(t)] = x(t)^\top X_{f_k} x(t)$ constructed using the solution X_{f_k} of (28), (45), respectively (51).

Comment on the Robust Stability Bound III

Above, the spectral-norm $\|X\|_S$ is the maximum singular value of matrix X

$$\|X\|_S = \sigma_{\max}(X) = \sqrt{\lambda_{\max}(X^\top X)}\,,$$

and, when X is a symmetric-matrix, one has

$$\|X\|_S = \sigma_{\max}(X) = \max\,|\lambda(X)|\,,$$

where $|\lambda(X)|$ denotes the absolute value of $\lambda(X)$.

Comment on the Robust Stability Bound IV

However, when X is a symmetric positive-definite matrix, one has

$$\|X\|_S = \sigma_{\max}(X) = \sqrt{\lambda_{\max}(X^\top X)} = \lambda_{\max}(X)\,,$$
$$\|X^{-1}\|_S = \lambda_{\max}(X^{-1}) = \frac{1}{\lambda_{\min}(X)}\,,$$

which transforms μ_{f_k} from (56) into

$$\mu_{f_k} = \frac{1}{2}\frac{\lambda_{\min}(T_{f_k})}{\lambda_{\max}(X_{f_k})} + \alpha_f \frac{\lambda_{\min}(X_{f_k})}{\lambda_{\max}(X_{f_k})}\,, \tag{58}$$

after taking into account that both X_{f_k}, T_{f_k} are symmetric positive-definite matrices.

Comment on the Robust Stability Bound V

This far, note that, if the pair (A_f, B_f) is identified correctly, then the post-fault system is indeed stabilized under the nominal stability-margin constraint α_f by the designed control and no supplementary condition on (A_f, B_f) is required.

However, regarding the robustness of the accommodated control, consider the uncertainties in the identification / estimation of the post-fault system, i.e., (A_f, B_f) are not known exactly, that is (A_f^*, B_f^*) represent the actual faulty system, while (A_f, B_f) stand for the model established by the FDIE algorithm; in this respect, there is to see under which conditions on the identification / estimation error the real post-fault system is indeed stabilized under the post-fault stability-margin constraint α_f by the accommodated control or not.

Comment on the Robust Stability Bound VI

For example, if the post-fault pair defines the system structured identification error fault and the loss-of-effectiveness actuator fault (A_f, B_f) identified / estimated by the FDIE module as

$$A_f = (1 - \upsilon_{af})A_f^* = A_f^* - \upsilon_{af}A_f^* \Rightarrow \Delta A_f = \upsilon_{af}A_f^* \,, \tag{59a}$$

$$B_f = (1 - \upsilon_{bf})B_f^* = B_f^* - \upsilon_{bf}B_f^* \Rightarrow \Delta B_f = \upsilon_{bf}B_f^* \,, \tag{59b}$$

then $\epsilon_{\Delta A_f}$ respectively $\epsilon_{\Delta B_f}$ are computed as

$$\epsilon_{\Delta A_f} = \upsilon_{af}\sigma_{\max}(A_f^*) \,, \tag{60a}$$

$$\epsilon_{\Delta B_f} = \upsilon_{bf}\sigma_{\max}(B_f^*) \,, \tag{60b}$$

Comment on the Robust Stability Bound VII

from where (55c) with (58) gives the interdependent inequality condition on the maximum performance degradation encompassed by υ_{af} respectively υ_{bf} as

$$\upsilon_{af}\sigma_{\max}(A_f^*) + \sqrt{\frac{m}{n}}\upsilon_{bf}\sigma_{\max}(B_f^*)\sigma_{\max}(K_{f_k}) \leq \frac{1}{n}\mu_{f_k}\,, \tag{61}$$

and, if one of the two values either υ_{af} or υ_{bf} is fixed, then the other can be computed as

$$\upsilon_{af} \leq \frac{1}{\sigma_{\max}(A_f^*)}\left[\frac{1}{n}\mu_{f_k} - \sqrt{\frac{m}{n}}\upsilon_{bf}\sigma_{\max}(B_f^*)\sigma_{\max}(K_{f_k})\right]\,, \tag{62}$$

respectively

$$\upsilon_{bf} \leq \sqrt{\frac{n}{m}}\frac{1}{\sigma_{\max}(B_f^*)\sigma_{\max}(K_{f_k})}\left[\frac{1}{n}\mu_{f_k} - \upsilon_{af}\sigma_{\max}(A_f^*)\right]\,. \tag{63}$$

Outline

1. Optimal Model Matching
2. Modified Pseudo-Inverse Method – Revisited
   - Preliminaries
3. B747 Application Example

Preliminaries I

Unfortunately, remember that for MPIM there is always the drawback represented by the lack of closed-form solution for multi-input multi-output (MIMO) systems, that is the result from (64), respectively

$$k_{f_{i,j}}^{MPIM} = k_{f_{i,j}}^{PIM}\,, \text{ if } |k_{f_{i,j}}^{PIM}| \leq \tilde{\delta}_f\,, \text{ and}\,, \tag{64a}$$

$$k_{f_{i,j}}^{MPIM} = \text{sign}\left(k_{f_{i,j}}^{PIM}\right)\tilde{\delta}_f\,, \text{ otherwise}\,, \tag{64b}$$

gives an explicit solution for the post-fault control matrix when the plant is a single-input system ($B_f \in \mathbb{R}^{n\times 1}$) or it can be decomposed into such single-input subsystems.

δ-MPIM approximate control I

In the classical version of MPIM, for certain instability cases, it was suggested to use first an efficient computational method to get the stabilizing gain, for example, through solving a continuous algebraic Riccati equation (CARE)

$$CARE_f: \; A_f^\top X_f + X_f A_f - X_f B_f R_f^{-1} B_f^\top X_f + Q_f = 0\,, \tag{65}$$

with (almost) arbitrary penalty matrices $\bar{Q}_f^\top \bar{Q}_f = Q_f^\top = Q_f \geq 0$ and $R_f^\top = R_f > 0$, and then a correction to be made as follows

$$K_f^{\delta\text{-}MPIM} = K_f^{CARE} + \tilde{\delta}_f \Delta K_f^{PIM\text{-}CARE}\,, \tag{66}$$

where

$$K_f^{CARE} = R_f^{-1} B_f^\top X_f\,, \tag{67}$$

and the control accommodation difference $\Delta K_f^{PIM\text{-}CARE}$ is obtained as

$$\Delta K_f^{PIM\text{-}CARE} = B_f^\dagger \left[(A_f - B_f K_f^{CARE}) - \hat{A}_n^* \right]\,, \tag{68}$$

δ-MPIM approximate control II

where $\tilde{\delta}_f$ is the relaxed stability bound found for the restrictive one δ_f in the simplified expression of the impairment matrix $\tilde{E}_f$ given by

$$\tilde{E}_f = \delta_f B_f K_f^{PIM}\,, \tag{69}$$

following the control K_f^{PIM} obtained in (70), respectively

$$(K_f^{PIM} \equiv)\ K_f = B_f^{\dagger}(A_f - \hat{A}_n^*)\,, \tag{70}$$

and the bound δ_f to be found using (71), respectively

$$|c_{f_j}| < \frac{1}{\sigma_{\max}\left(\sum_{i=1}^{m} |\hat{X}_{f_i}|\right)}\,,\ \text{for } j = 1, 2, \ldots, m\,; \tag{71}$$

this approach might be called δ-MPIM control.

δ-MPIM approximate control III

However, in our approach, it is suggested to obtain the post-fault control matrix K_f^{GCARE} as

$$K_{f_k}^{GCARE} = R_{f_k}^{-1}(B_f^\top X_{f_k} + L_{f_k}), \tag{72}$$

through solving the GCARE$_{f_k}$'s associated with the faulty pair (A_f, B_f), that is (28), (45), and (51), respectively, precisely

$$GCARE_{f_k} \; : \; \tilde{A}_{f_k}^\top X_{f_k} + X_{f_k}\tilde{A}_{f_k} - X_{f_k}B_f R_{f_k}^{-1} B_f^\top X_{f_k} + \tilde{Q}_{f_k} = 0, \tag{73}$$

after an appropriate choice of the penalty matrices $Q_{f_k}, R_{f_k}, L_{f_k}$ from (30), (47), and (53), respectively, afterwards replacing K_f^{CARE} from (66)-(68) with $K_{f_k}^{GCARE}$ from (27), (44), and (50), respectively.

γ-MPIM approximate control I

Furthermore, in the classical version of MPIM, it was mentioned that the performance can also be improved by tuning both the stabilizing control gain K_f^{CARE} and the difference $\Delta K_f^{PIM\text{-}CARE}$ as

$$K_f^{\gamma\text{-}MPIM} = (1-\gamma_f)K_f^{CARE} + \gamma_f \Delta K_f^{PIM\text{-}CARE}; \tag{74}$$

in (74), the parameter γ_f is increased to the largest number smaller than one before the system becomes unstable; this approach might be called γ-MPIM control.

As above, in our approach, K_f^{CARE} should be replaced by $K_{f_k}^{GCARE}$. However, observe that (74) stands for a bi-objective convex optimization problem that increases the computational complexity of the design.

In the sequel, the optimal model matching approaches are applied to design a fault tolerant control that addresses the structural fault in the model of the short-period (SP) mode of a B747 aircraft respectively the loss-of-effectiveness in the elevator actuator.

Outline

1. Optimal Model Matching
2. Modified Pseudo-Inverse Method – Revisited
3. B747 Application Example
   - Robust Optimal Model Matching - Structural Fault
   - Classical, Simplified and Extended ROMM Control
   - Closed-Loop Simulation

Robust Optimal Model Matching - Structural Fault I

In what follows, it is proposed to approaching the post-fault closed-loop to the nominal closed-loop performance by minimizing the quadratic criterion $J_{f_c}(\alpha_f, \hat{A}_n^*)$ from (26) and thus the general weighting matrices Q_{f_c}, R_{f_c}, and L_{f_c} are derived at instant t_f as follows

$$Q_{f_c} = \begin{bmatrix} 0.0000 & -0.0028 \\ -0.0028 & 5.3041 \end{bmatrix} \times 10^5 \,,$$
$$R_{f_c} = 80.8103 \,,$$
$$L_{f_c} = \begin{bmatrix} -8.0747 & -426.1568 \end{bmatrix} .$$

As the absolute value of the real-part of eigenvalues of $\hat{A}_n^*$ is 1.9580, any smaller α_f can be chosen to be $\alpha_f = 1.95$, and hence the accommodated post-fault closed-loop matrix is obtained, for

$$K_{f_c}^{ROMM} = \begin{bmatrix} -0.4865 & -77.4809 \end{bmatrix} ,$$

as follows

$$\hat{A}_{f_c}^{ROMM} = \begin{bmatrix} -3.9031 & 78.9341 \\ -0.2826 & -44.5042 \end{bmatrix} ,$$

with stable spectrum

$$\Lambda\left(\hat{A}_{f_c}^{ROMM}\right) = \{-4.4602, -43.9471\} \,,$$

Robust Optimal Model Matching - Structural Fault II

and remark that the complex-conjugateness of the poles-pair is lost.

Not much more is obtained using afterwards the modified pseudo-inverse adjustment as indicated by (66)-(69), precisely using the enhanced control gain matrix

$$K_{f_c}^{ROMM_a} = \begin{bmatrix} -0.4784 & -75.9653 \end{bmatrix},$$

for the difference

$$\Delta K_{f_c}^{ROMM} = \begin{bmatrix} 0.3866 & 72.2074 \end{bmatrix},$$

to obtain the closed-loop matrix

$$\hat{A}_{f_c}^{ROMM_a} = \begin{bmatrix} -3.8303 & 92.5309 \\ -0.2779 & -43.6269 \end{bmatrix},$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_c}^{ROMM_a}\right) = \{-4.4873, -42.9699\}$$

does not improve significantly.

Simplified Robust Optimal Model Matching – Structural Fault I

For the same fault as in the case of the classical Robust Optimal Model Matching, in what follows, it is proposed to minimizing the quadratic criterion $J_{f_s}(\alpha_f, \hat{A}_n^*)$ from (43) and thus the weighting matrices Q_{f_s}, R_{f_s}, and L_{f_s} are derived as follows

$$Q_{f_s} = \begin{bmatrix} 0.0008 & 0.0425 \\ 0.0425 & 2.2480 \end{bmatrix} \times 10^3 \,,$$
$$R_{f_s} = 80.8103 \,,$$
$$L_{f_s} = \begin{bmatrix} -8.0747 & -426.1568 \end{bmatrix} .$$

For the same $\alpha_f = 1.95$ as before, the accommodated post-fault closed-loop matrix is obtained, for

$$K_{f_s}^{ROMM} = \begin{bmatrix} -0.0089 & -2.6415 \end{bmatrix} ,$$

as follows

$$\hat{A}_{f_s}^{ROMM} = \begin{bmatrix} 0.3819 & 750.3040 \\ -0.0061 & -1.1834 \end{bmatrix} ,$$

with stable spectrum

$$\Lambda\left(\hat{A}_{f_s}^{ROMM}\right) = \{-0.4008 \pm 1.9959\jmath\} \,,$$

Simplified Robust Optimal Model Matching – Structural Fault II

and see that the complex-conjugateness of the poles-pair is regained.

Again, not much more is obtained using afterwards the modified pseudo-inverse adjustment, precisely using the enhanced control gain matrix

$$K_{f_s}^{ROMM_a} = \begin{bmatrix} -0.0108 & -2.6967 \end{bmatrix},$$

for the difference

$$\Delta K_{f_s}^{ROMM} = \begin{bmatrix} -0.0911 & -2.6321 \end{bmatrix},$$

to obtain the closed-loop matrix

$$\hat{A}_{f_s}^{ROMM_a} = \begin{bmatrix} 0.3647 & 749.8084 \\ -0.0072 & -1.2154 \end{bmatrix},$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_s}^{ROMM_a}\right) = \{-0.4253 \pm 2.1906\jmath\}$$

does not change dramatically.

Extended Robust Optimal Model Matching – Structural Fault I

For the same fault as in the case of the classical Robust Optimal Model Matching, in what follows, it is proposed to minimizing the quadratic criterion $J_{f_e}(\alpha_f, \hat{A}_n^*)$ from (49) and thus the weighting matrices Q_{f_e}, R_{f_e}, and L_{f_e} are derived as follows

$$Q_{f_e} = \begin{bmatrix} 0.0008 & 0.0426 \\ 0.0426 & 2.2549 \end{bmatrix} \times 10^3 \,,$$
$$R_{f_e} = 81.8103 \,,$$
$$L_{f_e} = \begin{bmatrix} -8.0817 & -428.7908 \end{bmatrix} .$$

For the same $\alpha_f = 1.95$ as before, the accommodated post-fault closed-loop matrix is obtained, for

$$K_{f_e}^{ROMM} = \begin{bmatrix} -0.0108 & -3.0953 \end{bmatrix} ,$$

as follows

$$\hat{A}_{f_e}^{ROMM} = \begin{bmatrix} 0.3645 & 746.2326 \\ -0.0072 & -1.4461 \end{bmatrix} ,$$

with stable spectrum

$$\Lambda\left(\hat{A}_{f_e}^{ROMM}\right) = \{-0.5408 \pm 2.1418\jmath\} \,,$$

Extended Robust Optimal Model Matching – Structural Fault II

and see that the complex-conjugateness of the poles-pair is once again recovered compared to the classical version.

Still, not much is obtained using afterwards the modified pseudo-inverse adjustment, precisely using the enhanced control gain matrix

$$K_{f_e}^{ROMM_a} = \begin{bmatrix} -0.0127 & -3.1410 \end{bmatrix},$$

for the difference

$$\Delta K_{f_e}^{ROMM} = \begin{bmatrix} -0.0891 & -2.1782 \end{bmatrix},$$

to obtain the closed-loop matrix

$$\hat{A}_{f_e}^{ROMM_a} = \begin{bmatrix} 0.3477 & 745.8224 \\ -0.0083 & -1.4726 \end{bmatrix},$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_e}^{ROMM_a}\right) = \{-0.5624 \pm 2.3202\jmath\}$$

once more does not change much.

Extended Robust Optimal Model Matching – Structural Fault III

However, this is justified by the fact that, in the robust optimal model matching, the post-fault control matrix in (27) is composed of two parts, precisely $R_{f_c}^{-1} B_f^\top X_{f_c} + R_{f_c}^{-1} L_{f_c}$, of which the first component is the classical optimal control gain that minimizes a quadratic integral functional with Q_{f_c} and R_{f_c} symmetric weighting matrices, but from the identification of L_{f_c} as $B_f^\top (A_f - \hat{A}_n^*)$ see that the second component is nothing that K_f^{PIM} (for $m = 1$).

Classical, Simplified and Extended ROMM Control I

Furthermore, from a model matching perspective, see that the cost (75), respectively

$$J_f(K_f) = \left\| (A_f - B_f K_f) - \hat{A}_n^* \right\|_F^2, \tag{75}$$

is more linked with (43) and (49) than with (26), as the simplified and the extended ROMM controls provide the difference from the reference model evaluated according to (75) as

$$J_f\left(K_{f_s}^{ROMM_a}\right) = \left\| \hat{A}_{f_s}^{ROMM_a} - \hat{A}_n^* \right\|_F^2 = 537.8310\,,$$
$$J_f\left(K_{f_e}^{ROMM_a}\right) = \left\| \hat{A}_{f_e}^{ROMM_a} - \hat{A}_n^* \right\|_F^2 = 368.7085\,,$$

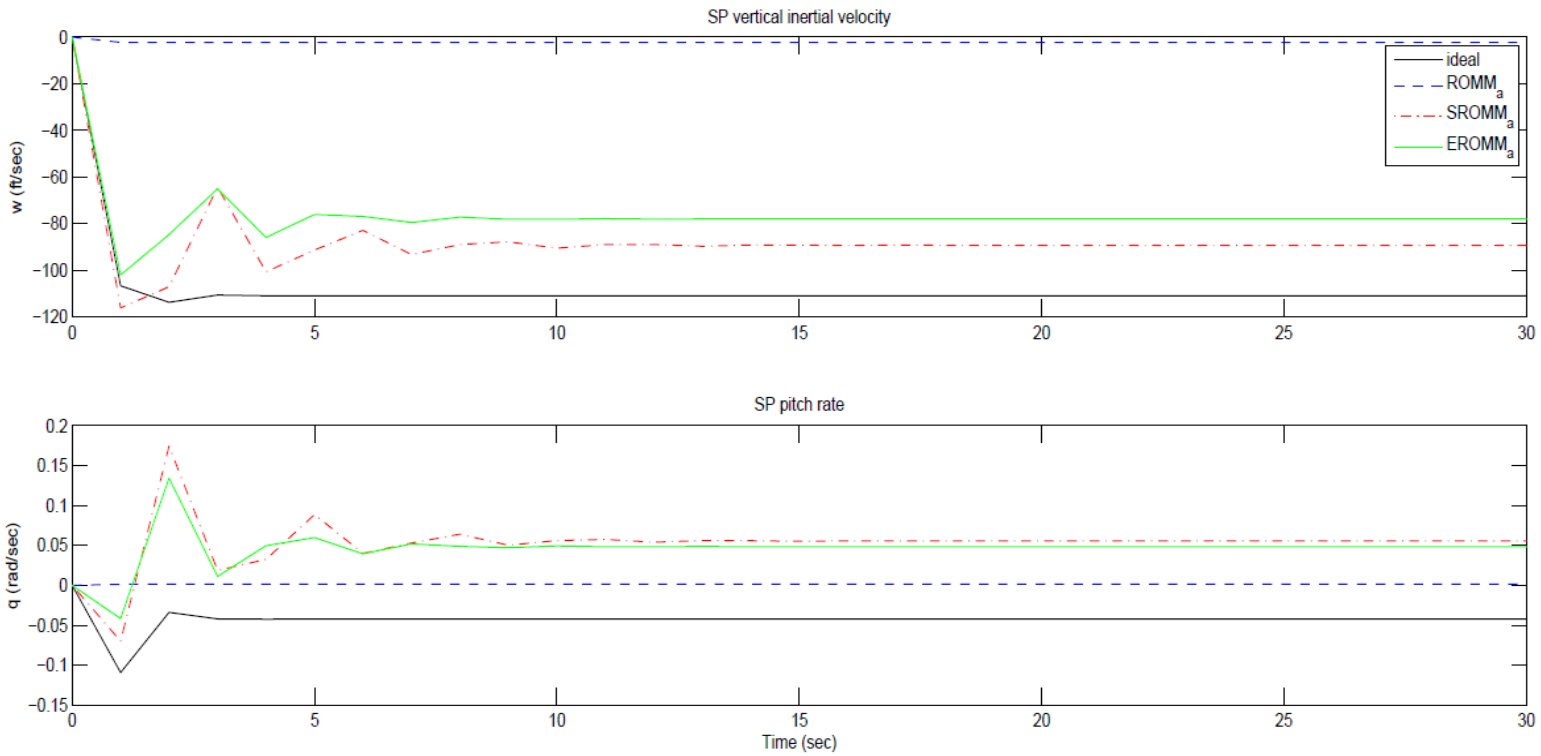
which are three orders of magnitude smaller than the one evaluated for the classical ROMM-control that provides

$$J_f\left(K_{f_c}^{ROMM_a}\right) = \left\| \hat{A}_{f_c}^{ROMM_a} - \hat{A}_n^* \right\|_F^2 = 4.0385 \times 10^5\,,$$

and thus the simplified and extended controls derived from the simplified and from the extended ROMM yield better performance than the classical control derived from the classical ROMM when the three approaches are used alternatively in the MPIM with correction.

Closed-Loop Simulation I

The short-period mode behavior is shown in the following figure, during the post-fault period (restricted on the graph to $t_f = 0$), when applying the classical respectively the simplified and the extended ROMM accommodating controls.



Conclusion I

In these series of lectures, the nominal and fault tolerant model matching problems are described. Upon the classical pseudo- and modified pseudo-inverse methods, two fault accommodation strategies are proposed, namely the admissible and robust optimal model matching, in order to obtain the corresponding feedback laws for the passive respectively active treatments of system robustness to faults.

An application to the accommodation for the structural fault in the short-period mode model of the B747 aircraft and the loss-of-effectiveness of the actuator controlling the elevator deflection was also developed. Simulations reveal the fact that the robust optimal model matching avoids an ulterior adjustment with the modified pseudo-inverse as it uses the pseudo-inverse intrinsically.

Also, compared to the robust optimal model matching used in conjunction with the modified pseudo-inverse method, the simplified respectively extended robust optimal model matching versions proposed and used similarly as the classical method provide much better results from the model matching point of view than the first one.

Questions? vs. Answers!

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